

Abhishek Kumar

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PROFILE SUMMARY

Final-year B.Tech student at NIT Calicut with expertise in Data Structures, Algorithms, Object-Oriented Programming, and C++ systems development. Hands-on experience architecting network security tools (Deep Packet Inspection) and machine learning applications. Seeking a Software Development Engineer (SDE) role to build scalable, high-performance backend systems.

EDUCATION

National Institute of Technology, Calicut <i>B.Tech in Electrical and Electronics Engineering (CGPA: 7.22)</i>	Calicut, Kerala 2023 – 2027
Krishna Public School <i>Class 12 (CBSE) – Senior Secondary (Percentage: 90.4%)</i>	Patna, Bihar 2023
SKP Vidya Vihar <i>Class 10 (CBSE) – Secondary (Percentage: 91.3%)</i>	Deoghar, Jharkhand 2021

TECHNICAL SKILLS

Languages: C++, Python, JavaScript, HTML, CSS
CS Fundamentals: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks (TCP/IP)
Frameworks & Libraries: Scikit-Learn, NumPy, Pandas
Developer Tools: Git, GitHub, VS Code, MATLAB, Simulink, SPICE

PROJECTS

- Deep Packet Inspection (DPI) Engine** | *C++, PCAP, Computer Networks, Cybersecurity* Nov. 2025 – Jan. 2026
- Engineered a high-throughput C++ network packet inspection analyzer using libpcap for offline PCAP traffic analysis.
 - Parsed layered network protocols including Ethernet, IPv4, TCP, UDP, TLS, and QUIC to extract connection metadata and SNI attributes.
 - Implemented modular packet processing pipelines along with checksum validation algorithms to detect encrypted vs. unencrypted traffic anomalies.
 - Applied low-level socket and network layer concepts to classify connection states and monitor session statistics.
- Neural Network Spam Email Classifier** | *Python, Scikit-learn, NumPy, Pandas, OOP* Sep. 2025 – Oct. 2025
- Architected a custom backpropagation-based neural network leveraging OOP design patterns for modular and extensible model components.
 - Designed an end-to-end data processing pipeline incorporating text normalization, tokenization, and TF-IDF feature vectorization.
 - Tuned model hyperparameters and evaluated classification metrics (Precision, Recall, F1-score), optimizing test data generalization.

POSITIONS OF RESPONSIBILITY

- Operations & Logistics Lead** NIT Calicut
RAGAM Cultural Festival 2025 – 2026
- Managed end-to-end logistics, venue allocation, and crowd management strategies for campus events with over 1,000+ attendees.
 - Directed multi-departmental volunteer teams under tight deadlines to ensure zero execution delays.
- Creative Promotion Lead** NIT Calicut
Tathva Technical Fest 2024 – 2025
- Designed visual media assets and promotional campaign videos using digital editing tools, expanding official event reach on social platforms.